Direct requirements matrix (A matrix) and Total requirements matrix (L matrix)

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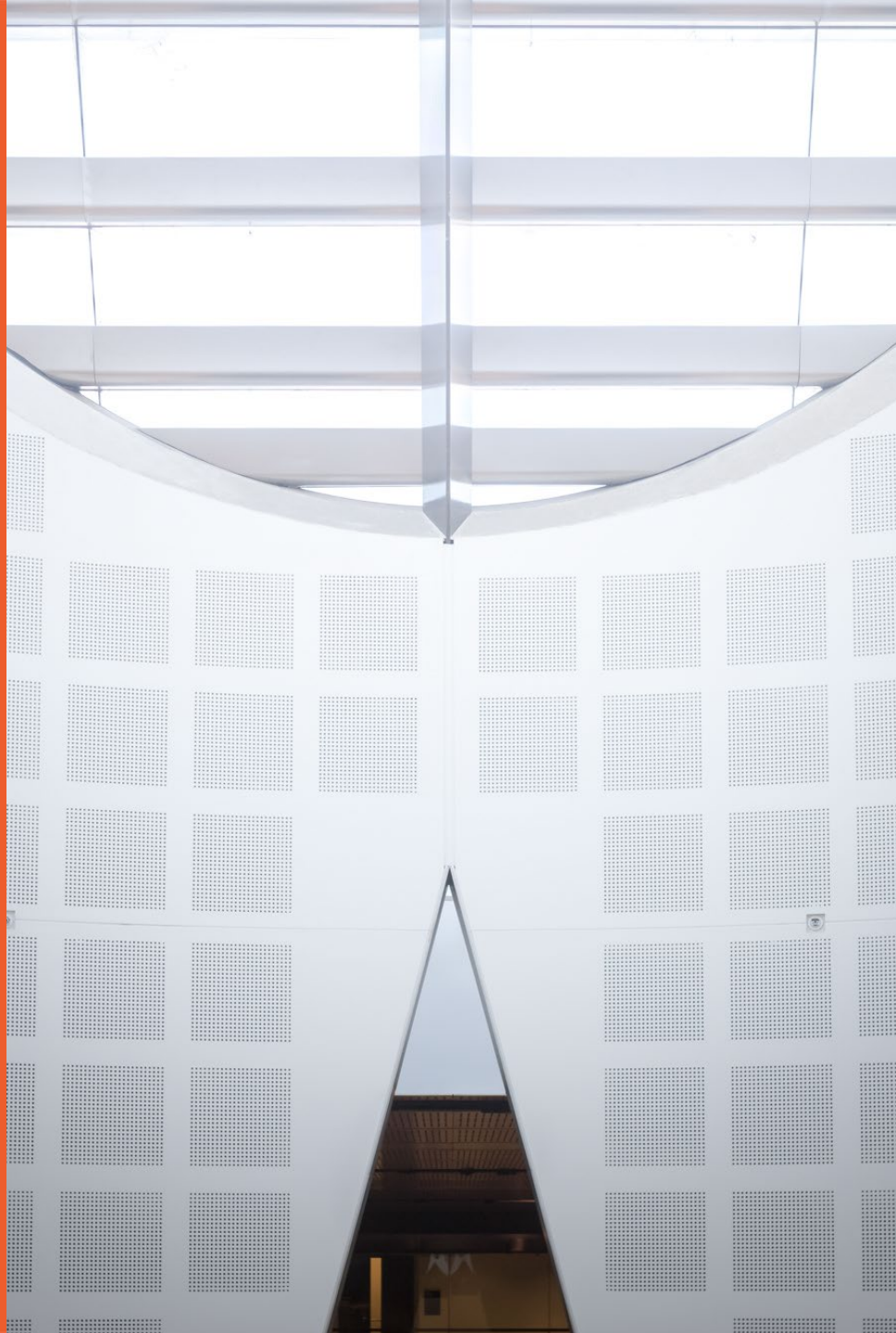
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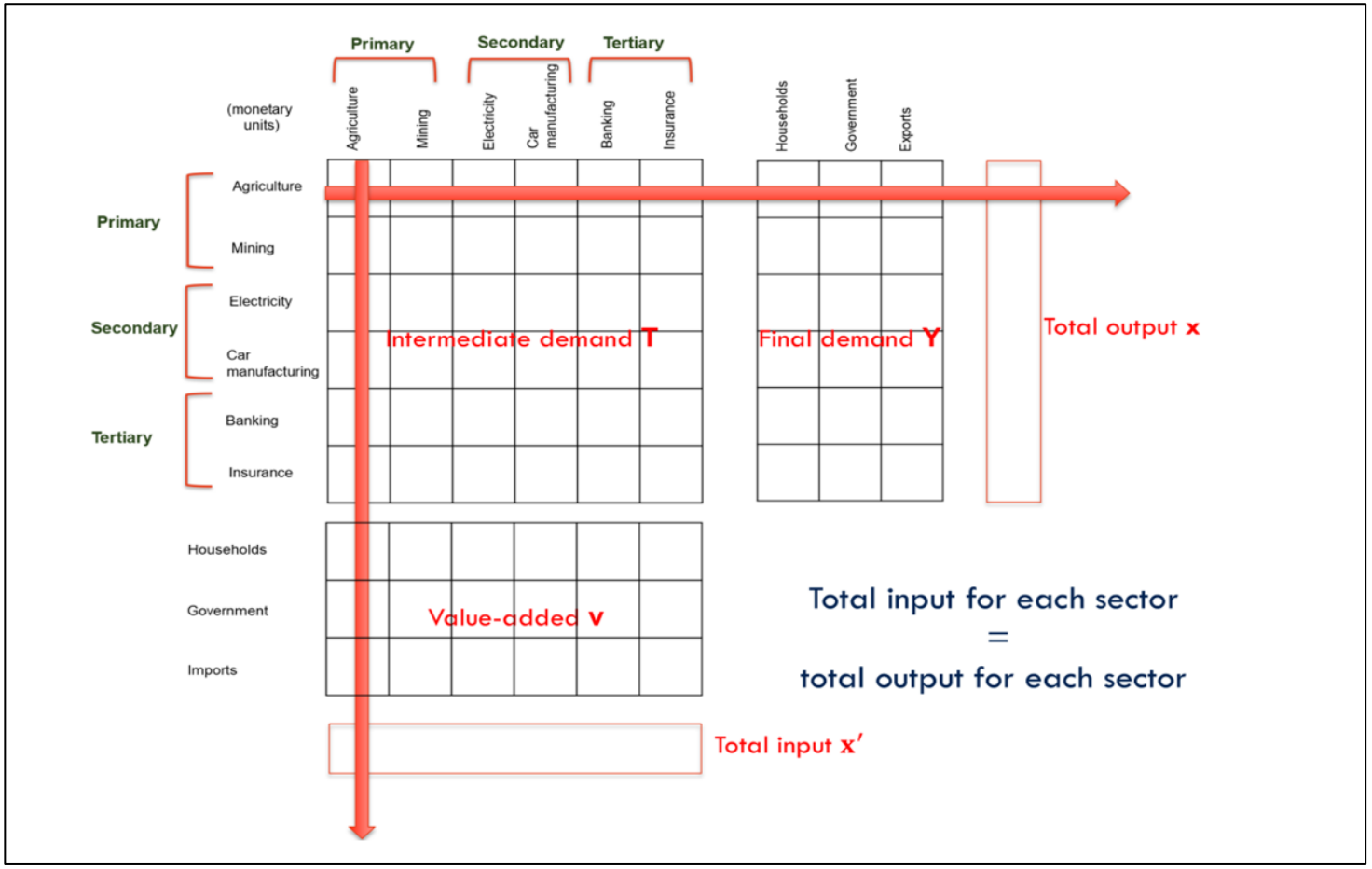


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Recap – Week 3

- Input-output table development
- Use of the System of National Accounts
- Basic price -> Producers' price -> Purchasers' price
- Difference between taxes and subsidies on production and taxes and subsidies on products

Recap thus far



What will we cover today?

Step 1 of calculating footprints; heavy on equations but easy to implement in Excel.

Do I need to remember all the derivations?

No, but it's good to know the 'path' to deriving the footprint equation.

Direct requirements matrix - A

- Also known as the direct coefficients matrix
- Provides us with information on the direct inputs required from every sector in order to produce $1 worth of output from a given sector *
- Dimensions are the same as the **T** matrix
- Calculated using values in **T** and the total output vector **x**

* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.

Direct requirements matrix - A

	A	F	S	H	G	E	x
A	5	6	7	2	2	1	23
F	5	2	1	1	0	5	14
S	6	3	4	1	2	0	16

H	2	1	1
I G	3	1	1
I	2	1	2

x'	23	14	16

3 intermediate demand sectors: Agriculture (A), Fishing (F), Services (S)
3 value-added entities: Households (H), Government (G), Imports (I)
3 final-demand entities: Households (H), Government (G), Exports (E)

Direct requirements matrix - A

Calculated using the values in **T** and **x**

	A	F	S
A	5	6	7
F	5	2	1
S	6	3	4

	H	G	E
A	2	2	1
F	1	0	5
S	1	2	0

	x
A	23
F	14
S	16

A matrix

A	5/23	6/14	7/16
F	5/23	2/14	1/16
S	6/23	3/14	4/16

H	2	1	1
I	3	1	1
G	2	1	2

H	2/23	1/14	1/16
I	3/23	1/14	1/16
G	2/23	1/14	2/16

x'	23	14	16

3 intermediate demand sectors: Agriculture (A), Fishing (F), Services (S)
3 value-added entities: Households (H), Government (G), Imports (I)
3 final-demand entities: Households (H), Government (G), Exports (E)

Direct requirements matrix - A

	A	F	S	H	G	E	x
A	5	6	7	2	2	1	23
F	5	2	1	1	0	5	14
S	6	3	4	1	2	0	16
H	2	1	1				
I	3	1	1				
G	2	1	2				
x'	23	14	16				

total output of the agricultural sector

A matrix

5/23	6/14	7/16
5/23	2/14	1/16
6/23	3/14	4/16
2/23	1/14	1/16
3/23	1/14	1/16
2/23	1/14	2/16

Production recipe
(Inputs needed to produce
$1 of output of the agricultural sector)

3 intermediate demand sectors: Agriculture (A), Fishing (F), Services (S)
3 value-added entities: Households (H), Government (G), Imports (I)
3 final-demand entities: Households (H), Government (G), Exports (E)

Direct requirements matrix - A

	A	F	S	H	G	E	x
A	5	6	7	2	2	1	23
F	5	2	1	1	0	5	14
S	6	3	4	1	2	0	16
H	2	1	1				
I	3	1	1				
G	2	1	2				
x'	23	14	16				

total output of the agricultural sector

A matrix

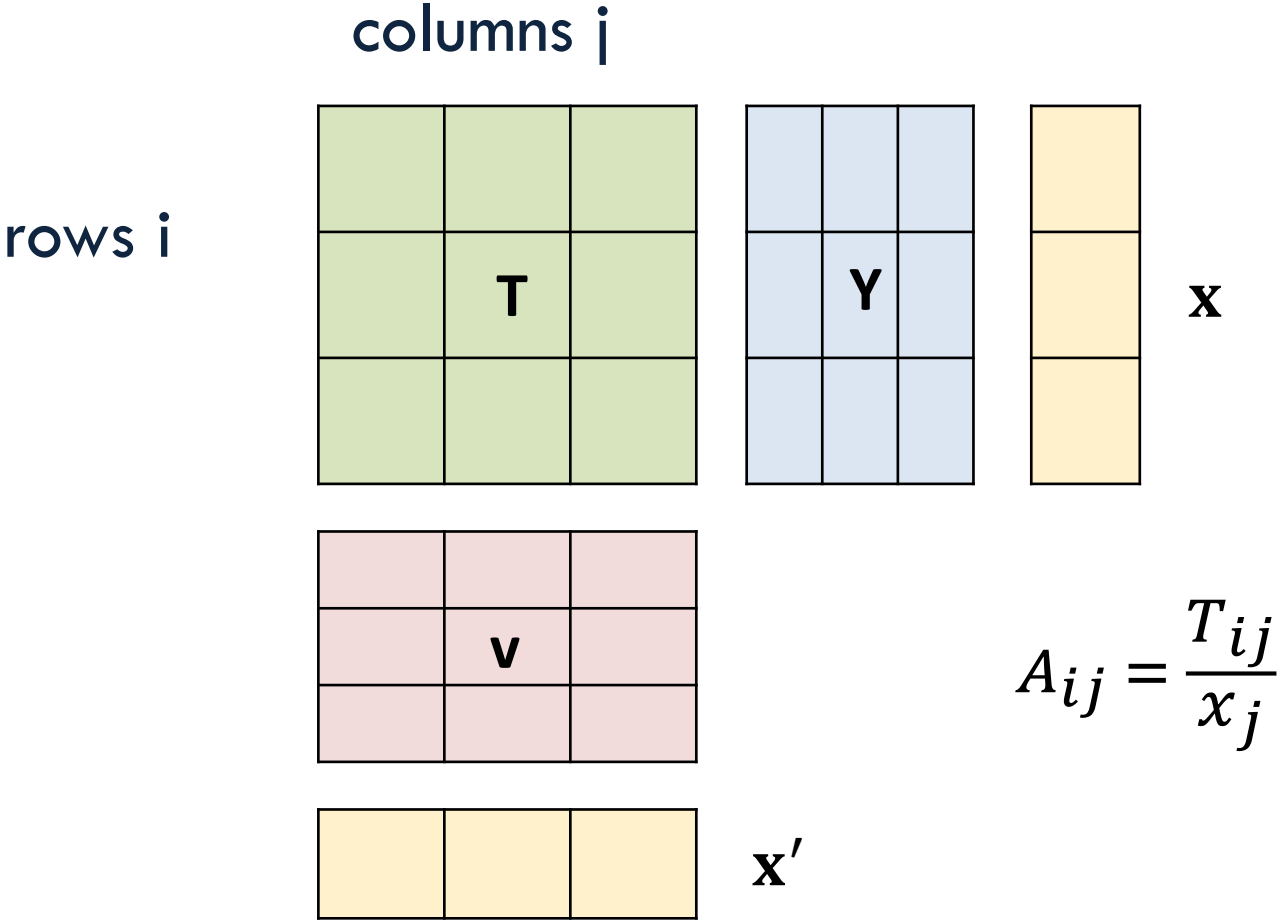
0.217	0.429	0.438
0.217	0.143	0.063
0.26	0.214	0.25
0.087	0.071	0.063
0.13	0.071	0.063
0.087	0.071	0.125

Production recipe
(Inputs needed to produce
$1 of output of the agricultural sector)

3 intermediate demand sectors: Agriculture (A), Fishing (F), Services (S)
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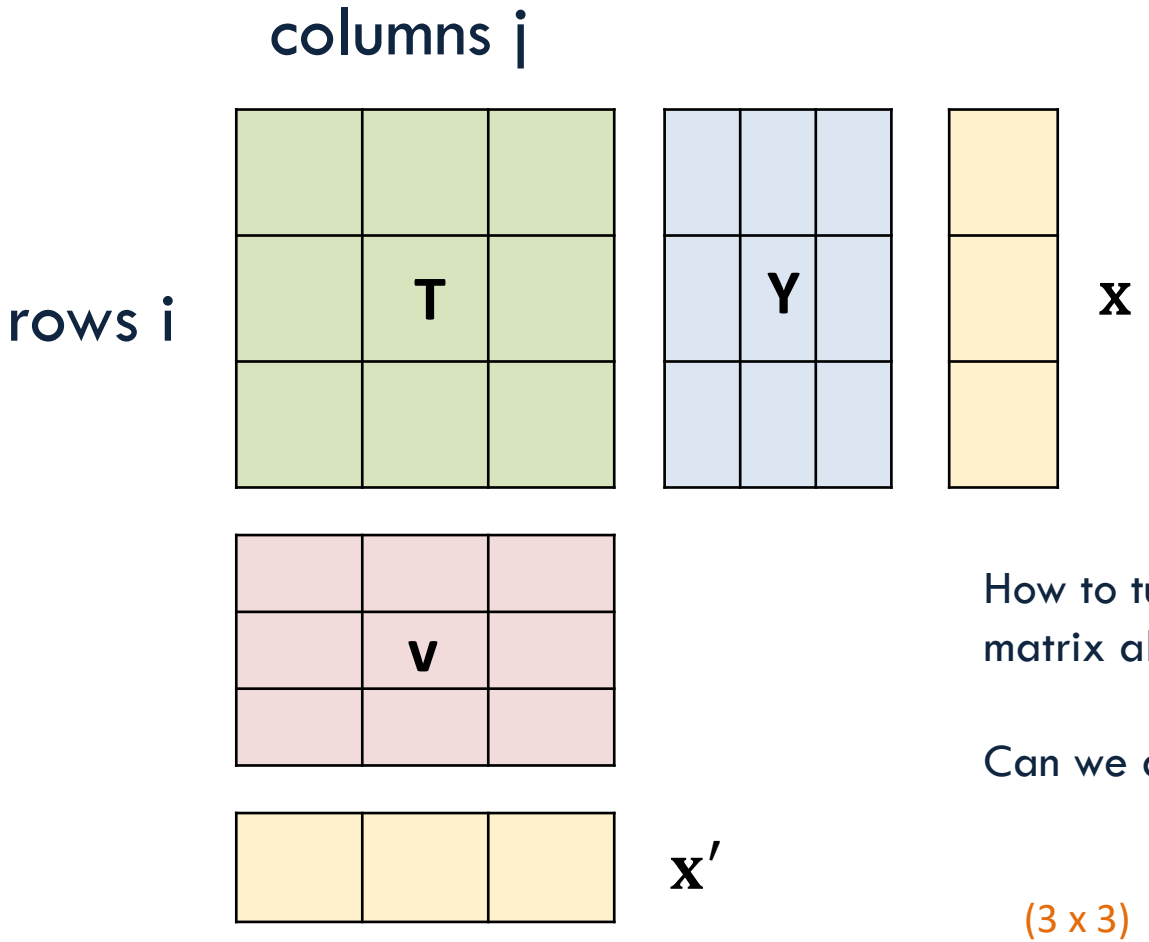
Direct requirements matrix - A

in terms of matrix notation…



$$A_{ij} = \frac{T_{ij}}{x_j}$$

In terms of matrix notation…



$$A_{ij} = \frac{T_{ij}}{x_j}$$

How to turn the **T** matrix into the **A** matrix all at once?

Can we do the following?

$$(3 \times 3) \quad A = \frac{T}{x} \quad \begin{matrix}(3 \times 3)\\(3 \times 1)\end{matrix}$$

This wouldn't work – we need **x** to have the same dimensions as **A** to provide an answer with dimensions equal to **T**.

Or, rearranging the equation:

$$(3 \times 3) \quad T = Ax$$

$$(3 \times 3) \quad (3 \times 1)$$

More fun with matrices

Diagonal matrices

Diagonal matrices are square matrices which contain zero values in all elements above and below the main diagonal.

1	0	0	0	0
0	7	0	0	0
0	0	23	0	0
0	0	0	4	0
0	0	0	0	67

Diagonal matrices

If $\mathbf{x} =$

1
7
23
4
67

then $\hat{\mathbf{x}} =$

1	0	0	0	0
0	7	0	0	0
0	0	23	0	0
0	0	0	4	0
0	0	0	0	67

"x hat"

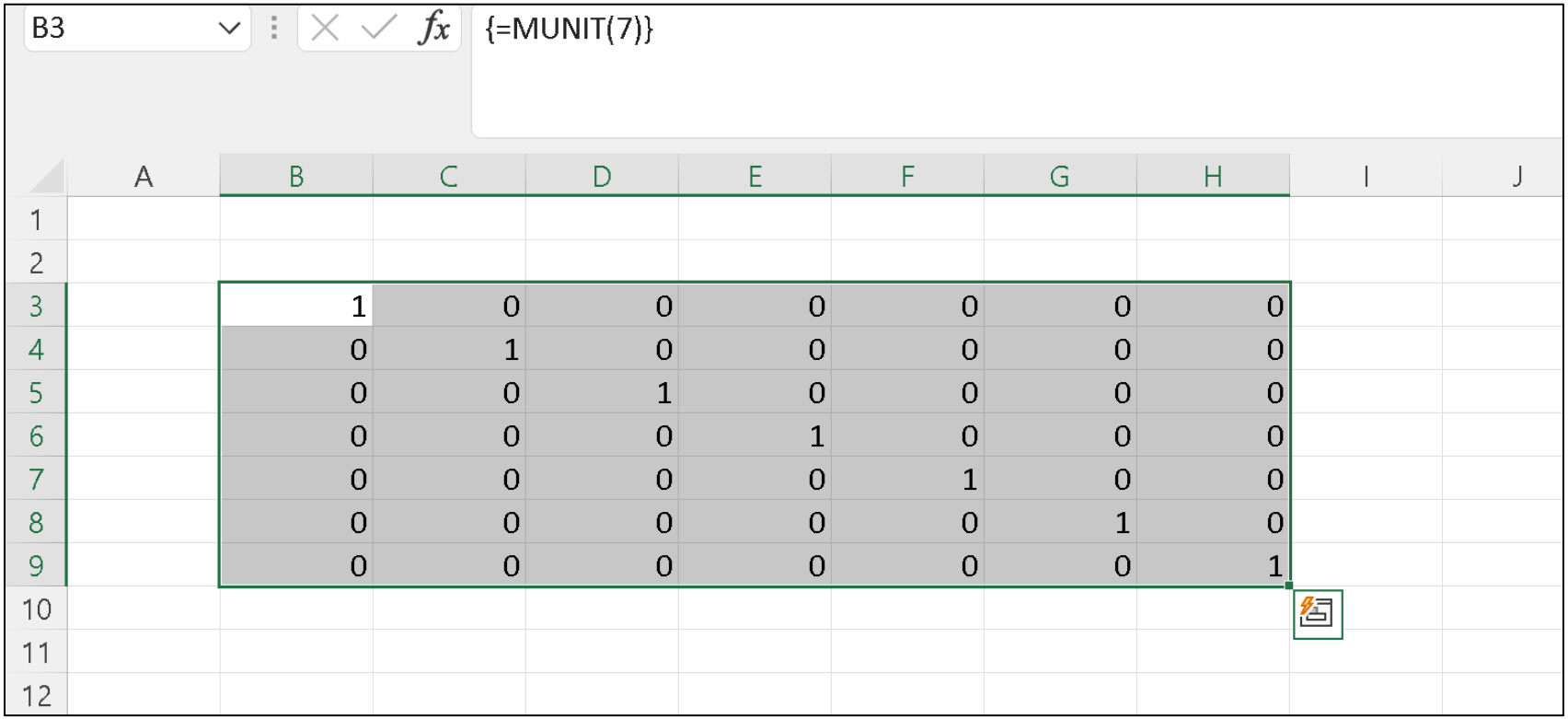
Identity matrix - I

A diagonal matrix with all elements on the diagonal equal to one (1) is known as the Identity matrix (**I**).

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

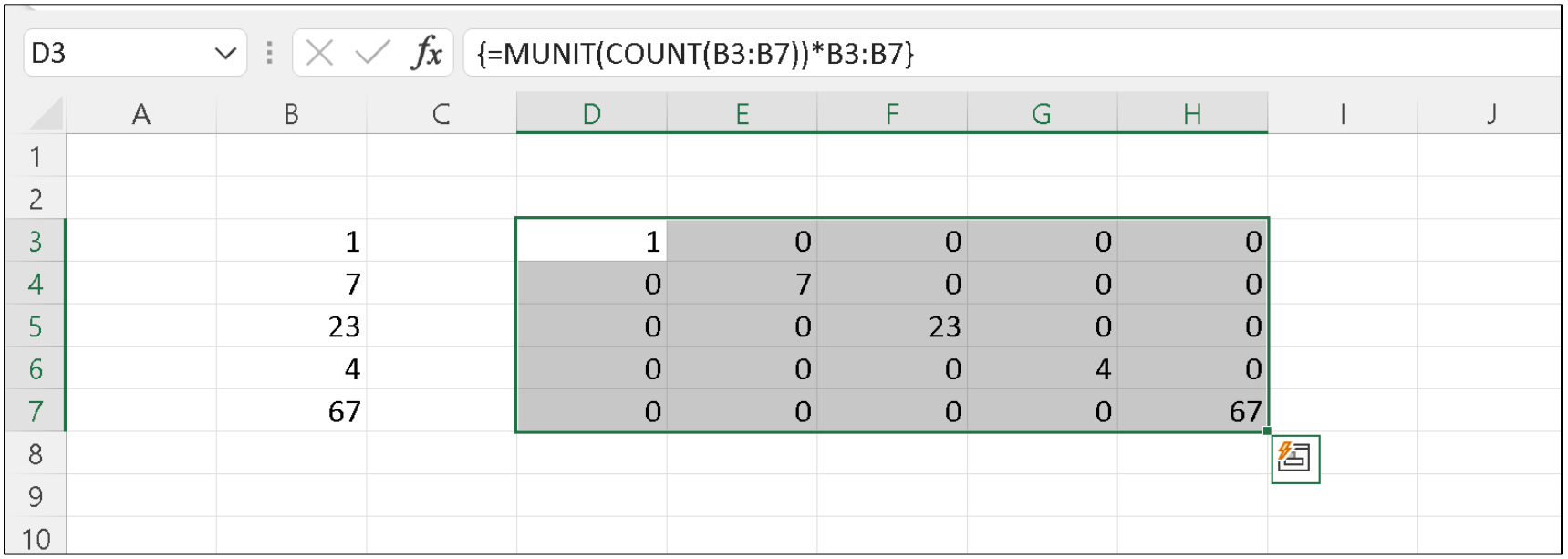
Using Excel to create the Identity matrix I

The MUNIT function in Excel allows you to easily create an identity matrix



Using Excel to create a diagonal matrix

The MUNIT function in Excel can also be used to diagonalise an existing matrix, in conjunction with the 'COUNT' operation:



	A	B	C	D	E	F	G	H	I	J
1										
2										
3		1		1	0	0	0	0		
4		7		0	7	0	0	0		
5		23		0	0	23	0	0		
6		4		0	0	0	4	0		
7		67		0	0	0	0	67		
8										
9										
10										

The inverse of a matrix

In 'normal' mathematics, the inverse of a number is calculated by dividing 1 by that number:

$\frac{1}{3}$ is the inverse of 3

$3 \times \frac{1}{3} = 1$

$\frac{1}{w}$ is the inverse of w

$w \times \frac{1}{w} = 1$

$\frac{1}{w}$ can also be written as w^{-1}

The inverse of a matrix 反矩陣

When working with matrices, any matrix multiplied by its inverse returns the Identity matrix (**I**)

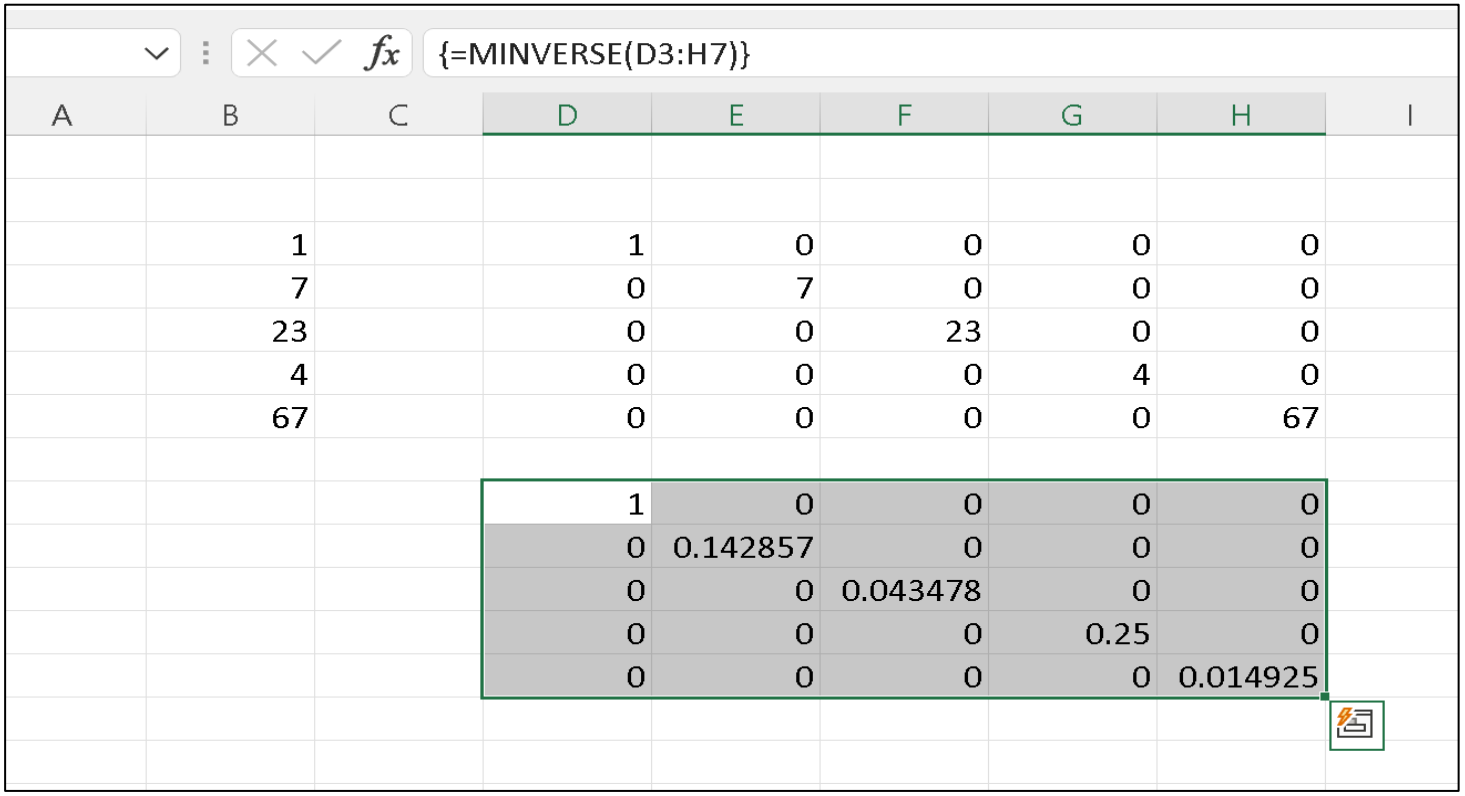
$$\mathbf{I} \times \mathbf{W} = \mathbf{W}$$

$$\mathbf{W} \times \frac{1}{\mathbf{w}} = \mathbf{I}$$

$$\mathbf{W} \times \mathbf{W}^{-1} = \mathbf{I}$$

Using Excel to create the inverse of a matrix

The MINVERSE function in Excel calculates the inverse of a matrix



Practice

1. Open up Excel and create a 3 x 3 matrix

1	0	1
2	3	2
4	3	2

2. Calculate the inverse of this matrix using MINVERSE

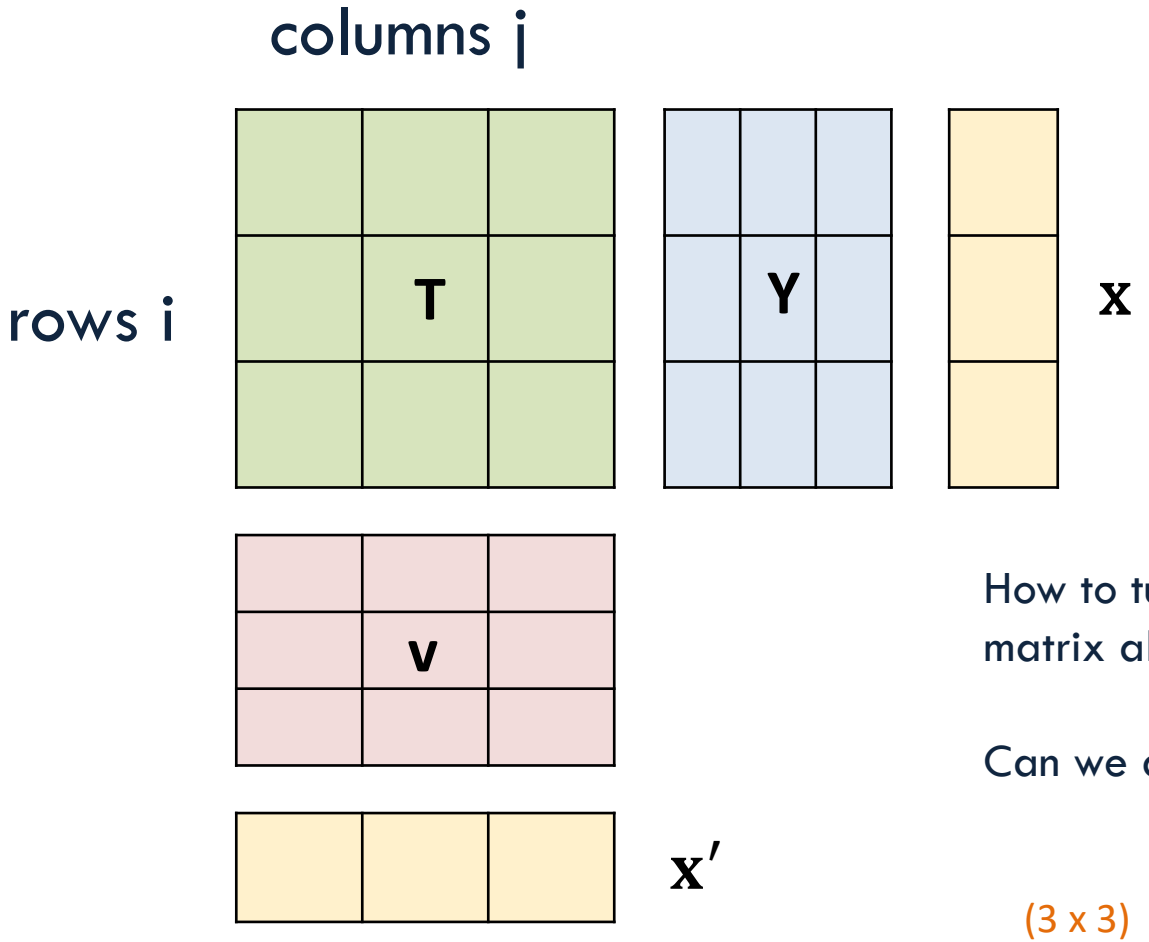
0	-0.5	0.5
-0.667	0.333	0
1	0.5	-0.5

3. Multiply the original matrix by its inverse using MMULT

4. Check to see if your answer is the identity matrix I

1	0	0
0	1	0
0	0	1

In terms of matrix notation…



$$A_{ij} = \frac{T_{ij}}{x_j}$$

How to turn the **T** matrix into the **A** matrix all at once?

Can we do the following?

$$(3 \times 3) \quad A = \frac{T}{x} \quad \begin{matrix}(3 \times 3)\\(3 \times 1)\end{matrix}$$

This wouldn't work – we need **x** to have the same dimensions as **A** to provide an answer with dimensions equal to **T**.

Or, rearranging the equation:

$$(3 \times 3) \quad T = Ax$$

$$(3 \times 3) \quad (3 \times 1)$$

Calculating the A matrix

$$\mathbf{T} = \mathbf{A}\mathbf{x}$$

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}}$$

Diagonalising **x** changes its dimensions, but not the values in it

How can we derive A?

$$\mathbf{A} = ?$$

Calculating the A matrix

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}}$$

What is the inverse of $\hat{\mathbf{X}}$? $\hat{\mathbf{X}}^{-1}$

Multiply both sides of the equation by $\hat{\mathbf{X}}^{-1}$

$$\mathbf{T}\hat{\mathbf{X}}^{-1} = \mathbf{A}\hat{\mathbf{X}}\hat{\mathbf{X}}^{-1}$$

$\hat{\mathbf{X}}\hat{\mathbf{X}}^{-1}$ → Identity matrix

Substitute $\hat{\mathbf{X}}\hat{\mathbf{X}}^{-1}$ for the Identity matrix **I**

$$\mathbf{T}\hat{\mathbf{X}}^{-1} = \mathbf{AI}$$

Ignore **I**
(just as we would ignore 1 in 'normal' maths)

$$\mathbf{T}\hat{\mathbf{X}}^{-1} = \mathbf{A}$$

$$\boxed{\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}}$$

Recap thus far – we've worked out the A matrix

	A	F	S
A	5	6	7
F	5	2	1
S	6	3	4

	H	G	E
A	2	2	1
F	1	0	5
S	1	2	0

	x
A	23
F	14
S	16

H	2	1	1
G	3	1	1
I	2	1	2

x'	23	14	16

A matrix

0.217	0.429	0.438
0.217	0.143	0.063
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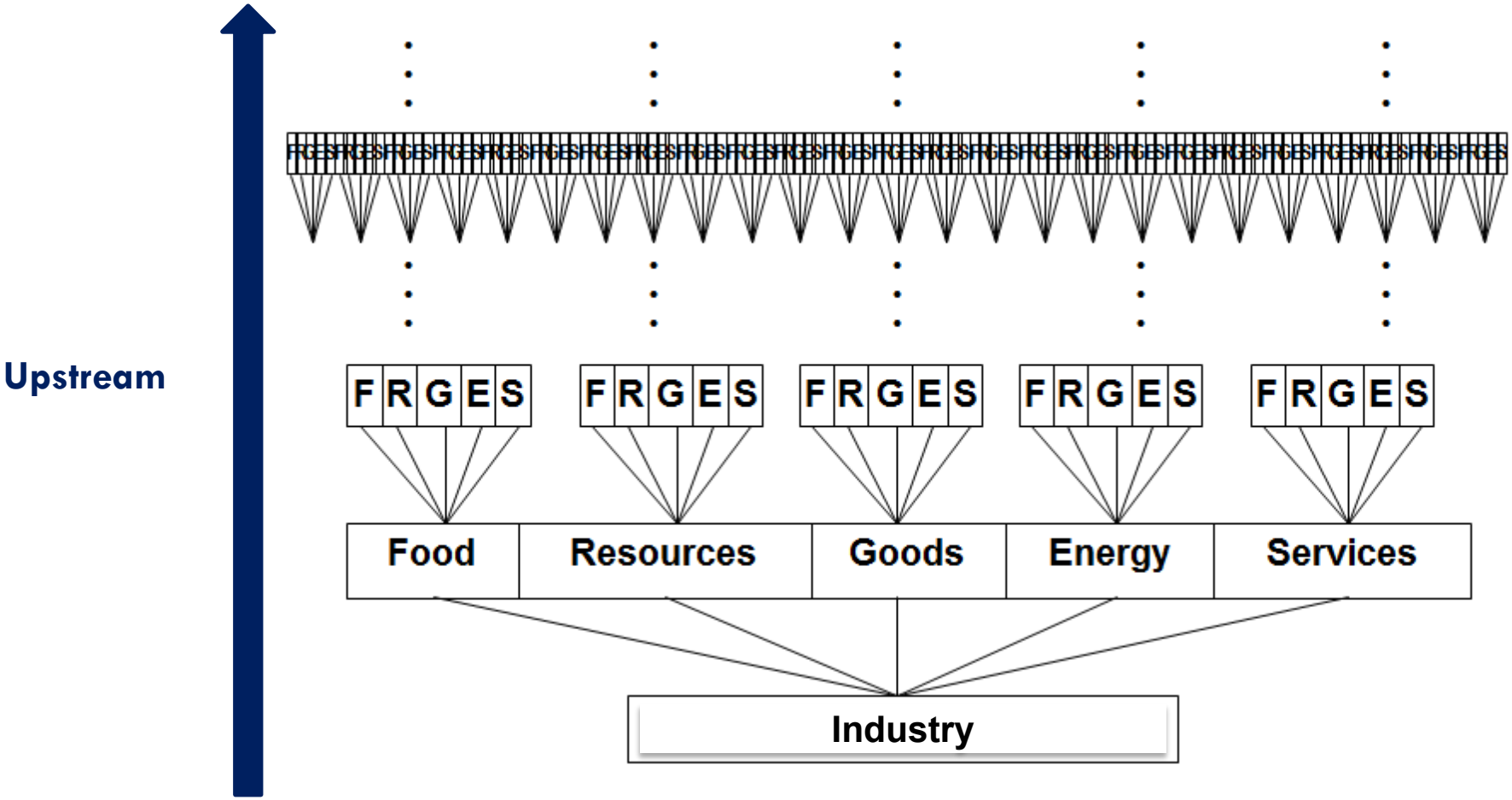
$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

$$A_{ij} = \frac{T_{ij}}{x_j}$$

3 intermediate demand sectors: Agriculture (A), Fishing (F), Services (S)
3 value-added sectors: Households (H), Government (G), Imports (I)
3 final-demand sectors: Households (H), Government (G), Exports (E)

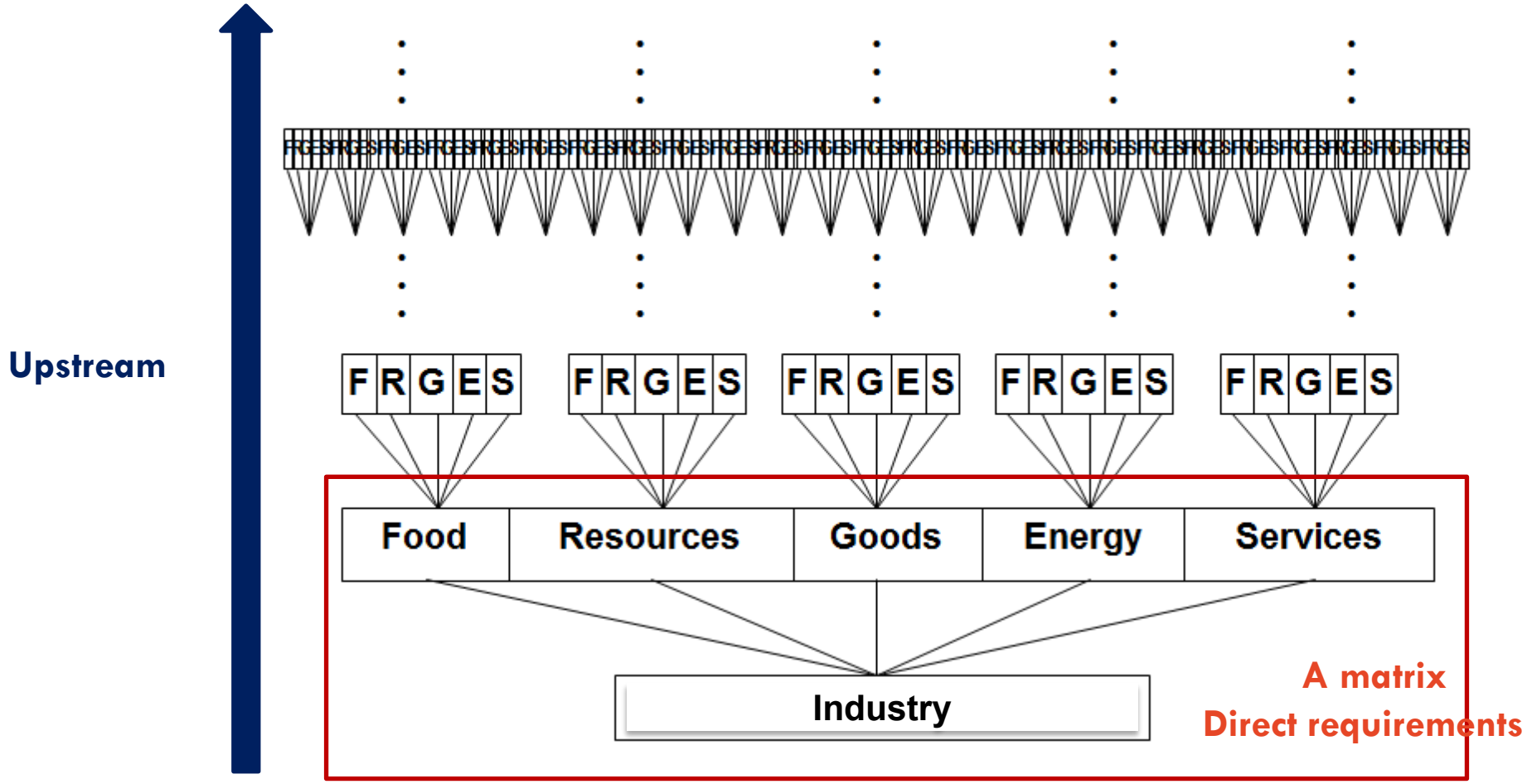
- **A** matrix – same size as the **T** matrix
- **A** matrix – direct requirements matrix/direct coefficients matrix
- **A** matrix – value of inputs directly required from each sector to provide one monetary unit worth of total output for a given sector

Do industries only rely on direct inputs?



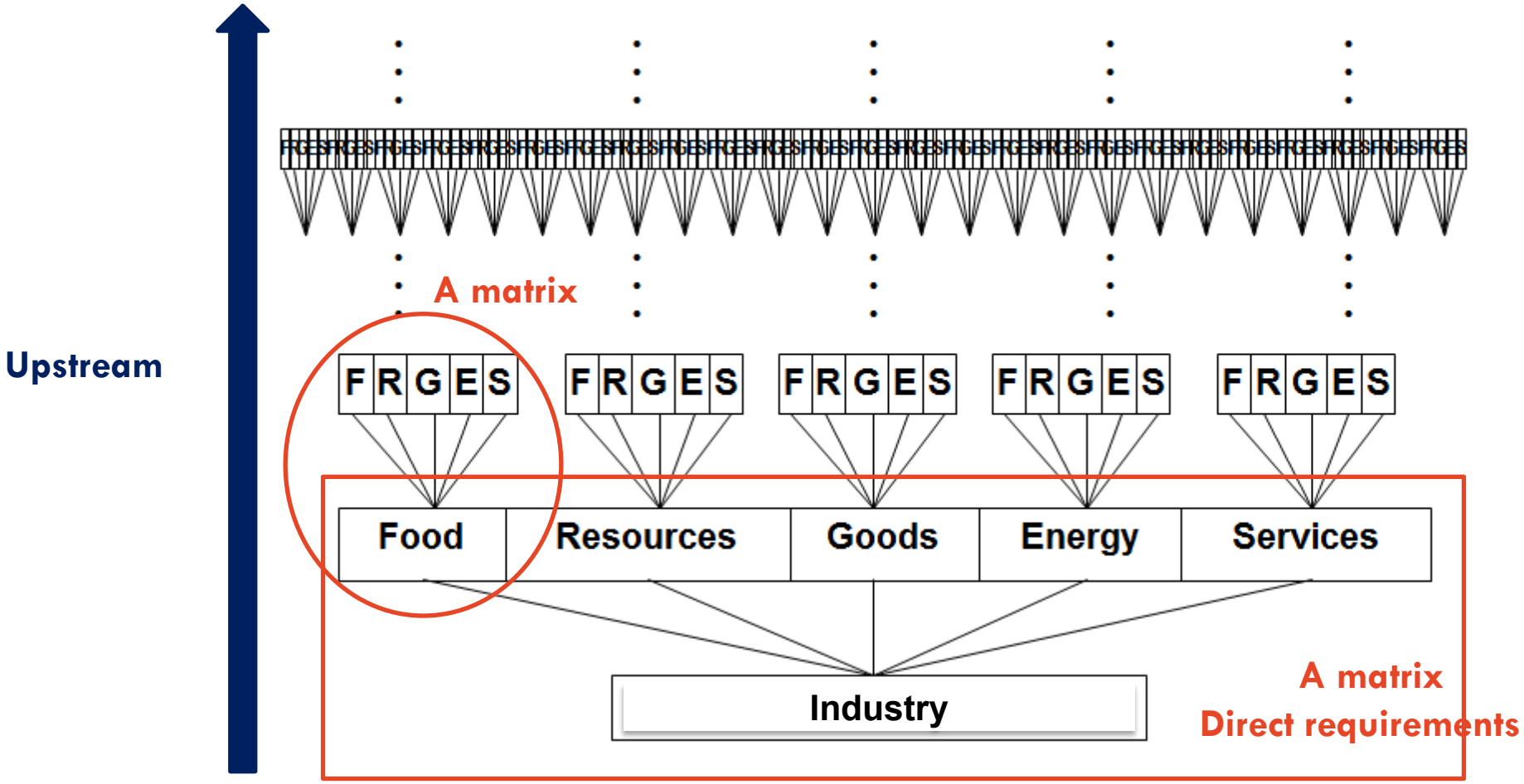
Do industries only rely on direct inputs?

We need to consider the supply chain!



Do industries only rely on direct inputs?

We need to consider the supply chain!



How can we quantify TOTAL (direct + indirect) requirements?

Recap of key equations

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}}$$

$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

Recap of key equations

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

Summation operator of T

Summation operator of y

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}}$$

Diagonal x

$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

Inverse of diagonal x

Recap of key equations

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}} \quad \text{——————} \quad 1)$$

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}} \quad \text{——————} \quad 2)$$

$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1} \quad \text{——————} \quad 3)$$

Deriving the fundamental Input-Output equation

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}} \qquad \text{1)}$$

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}} \qquad \text{2)}$$

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{x} = \mathbf{A}\hat{\mathbf{X}}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

Deriving the fundamental Input-Output equation

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}} \qquad \text{1)}$$

$$\mathbf{T} = \mathbf{A}\hat{\mathbf{X}} \qquad \text{2)}$$

Example: Imagine a 3 x 3 matrix

4	0	0	1	4
0	5	0	x 1 =	5
0	0	6	1	6
$\hat{\mathbf{X}}$			$1^{\mathbf{X}}$	$\mathbf{x}$

$$\mathbf{x} = \mathbf{T}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{x} = \mathbf{A}\hat{\mathbf{X}}1^{\mathbf{T}} + \mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{A}\mathbf{x} + \mathbf{Y}1^{\mathbf{Y}} = \mathbf{x}$$

Deriving the fundamental Input-Output equation

$$\mathbf{Ax} + \mathbf{Y1}^{\mathbf{Y}} = \mathbf{x}$$

Put x on one side

$$\mathbf{Y1}^{\mathbf{Y}} = \mathbf{x} - \mathbf{Ax}$$

Explicitly include **I** (just as we would include 1 in 'normal' maths)

$$\mathbf{Y1}^{\mathbf{Y}} = \mathbf{Ix} - \mathbf{Ax}$$

Factorise **x**

$$\mathbf{Y1}^{\mathbf{Y}} = (\mathbf{I} - \mathbf{A})\mathbf{x}$$

or

$$(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{Y1}^{\mathbf{Y}}$$

Deriving the fundamental Input-Output equation

$$(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{Y}1^{\mathbf{Y}}$$

Multiply both sides by the inverse of $(\mathbf{I} - \mathbf{A})$

$$(\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} - \mathbf{A})\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{Y}1^{\mathbf{Y}}$$

Recap: Any matrix multiplied by its inverse results in an identity matrix

$$\mathbf{I}\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{Y}1^{\mathbf{Y}}$$

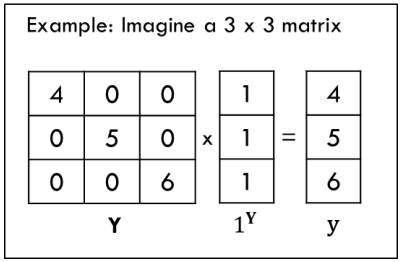
Recap: Any matrix/vector multiplied by an identity matrix results in the same matrix/vector

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{Y}1^{\mathbf{Y}}$$

Deriving the fundamental Input-Output equation

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{Y}1^{\mathbf{Y}}$$

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{y}$$



Total output — **L matrix** — **Total final demand**

Leontief inverse

$$\mathbf{x} = \mathbf{L}\boldsymbol{y}$$

Fundamental Input-Output Equation

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{y}$$

- Connects the total output of the economic system (**x**) to the consumption activity captured in the final demand vector (**y**).
- Once we have calculated **L**, we can connect any subset of consumption activity to the output of each sector required to satisfy that consumption activity.
- **L** provides us with information on the <u>total</u> (ie direct and indirect) requirements needed from each sector to provide $1 worth of total output for a given sector*.

* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.

Fundamental Input-Output Equation

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1} y$$

This relationship between total output and final demand relies on some assumptions (also see background reading material):

- Industries use the goods and services of other industries to produce their own goods or services, so the outputs of one industry are the inputs of another
- Transactions are accounted for in monetary terms
- There is a linear relationship between demand for inputs and outputs – if demand increases the output increases proportionately to meet the increased demand

Direct and total requirements matrices

A matrix - Direct requirements matrix

Provides information on the direct inputs required from each sector to produce $1 of output for a given sector. Every element of **A** will be less than 1*.

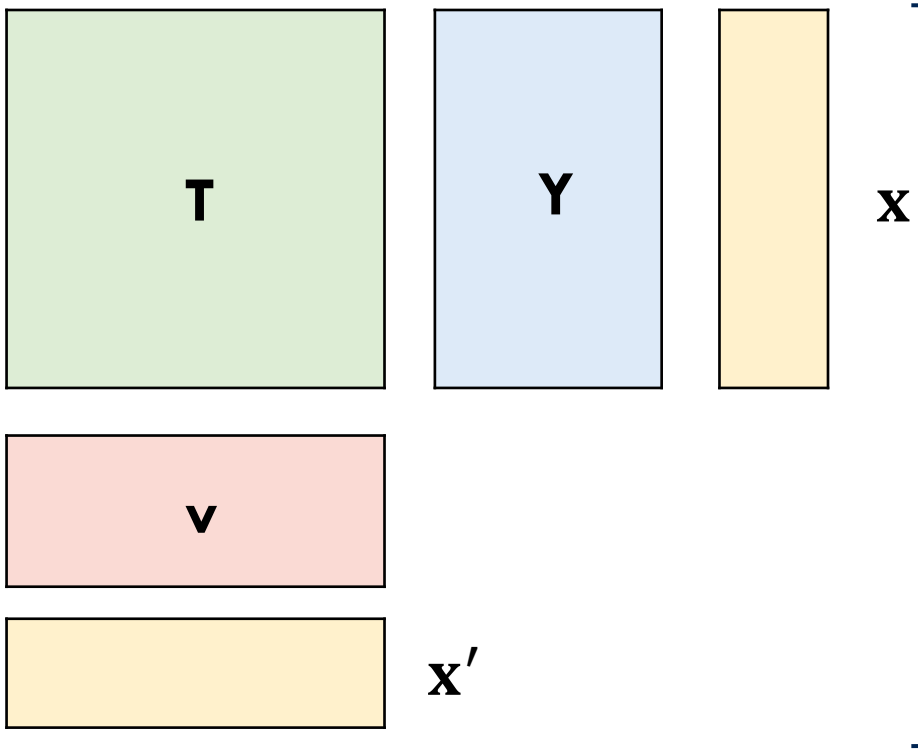
However, we know that each of the sectors that provide direct inputs into that given sector have their own direct requirements. The **A** matrix does not consider these indirect requirements, which represent the second and subsequent tiers of the original sector's supply chains.

L matrix – Leontief inverse or total requirements matrix

In order to understand both the direct and indirect requirements, we use the total requirements matrix (**L**), also known as the Leontief Inverse. This provides information on the total inputs required from each sector to produce $1 of output for a given sector. Every element within **L** will be greater than the element in the same location within **A**, and the elements on the diagonal of **L** will be greater than 1.

* It is mathematically possible for an element of A to be equal to 1 if there is no final demand for a sector, and the sector has no interaction with any other sectors, which is effectively impossible within an IO table.

Summary



$$A = T\hat{X}^{-1}$$

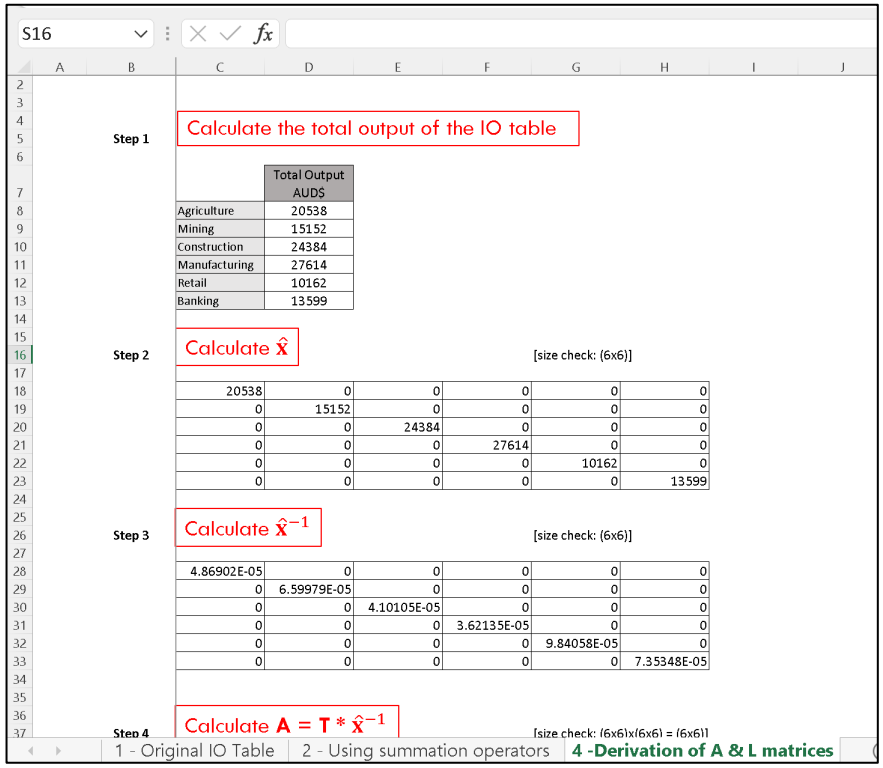
A

$$L = (I - A)^{-1}$$

- Create an identity matrix (same size as the **A** matrix)
- Calculate **I – A**
- Calculate the inverse of **I – A** using MINVERSE

- **T** matrix; **A** matrix; **L** matrix – SAME SIZE!
- **A** matrix – Direct requirements matrix
- **L** matrix – Total requirements matrix

Worked example on Canvas



S16

Step 1 Calculate the total output of the IO table

	Total Output AUD$
Agriculture	20538
Mining	15152
Construction	24384
Manufacturing	27614
Retail	10162
Banking	13599

Step 2 Calculate $\hat{\mathbf{x}}$ [size check: (6x6)]

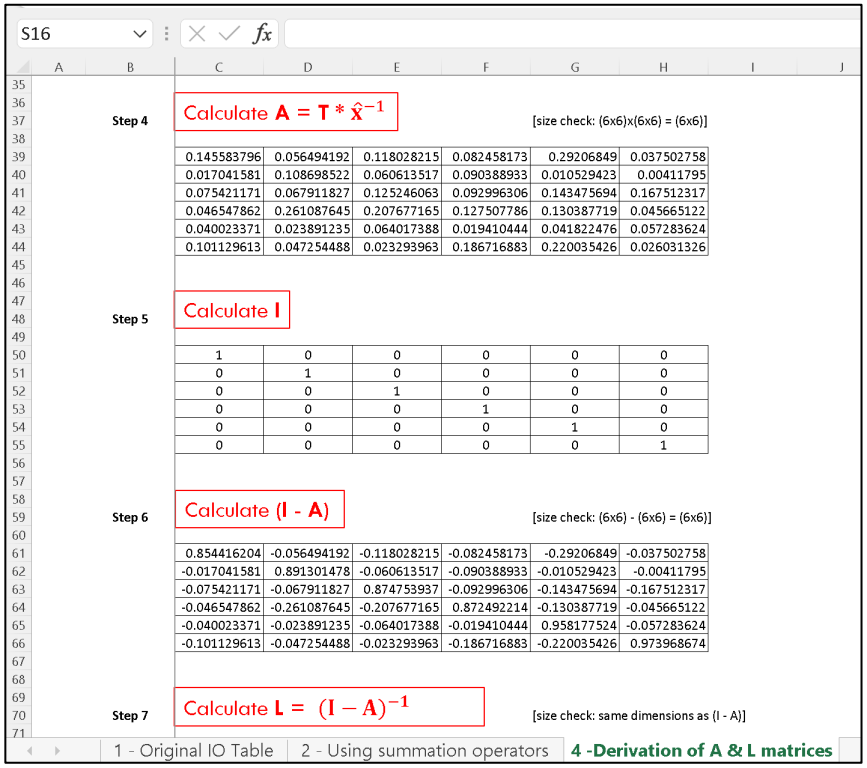
20538	0	0	0	0	0
0	15152	0	0	0	0
0	0	24384	0	0	0
0	0	0	27614	0	0
0	0	0	0	10162	0
0	0	0	0	0	13599

Step 3 Calculate $\hat{\mathbf{x}}^{-1}$ [size check: (6x6)]

4.86902E-05	0	0	0	0	0
0	6.59979E-05	0	0	0	0
0	0	4.10105E-05	0	0	0
0	0	0	3.62135E-05	0	0
0	0	0	0	9.84058E-05	0
0	0	0	0	0	7.35348E-05

Step 4 Calculate $\mathbf{A} = \mathbf{T} * \hat{\mathbf{x}}^{-1}$ [size check: (6x6)x(6x6) = (6x6)]

1 - Original IO Table | 2 - Using summation operators | **4 -Derivation of A & L matrices**



S16

Step 4 Calculate $\mathbf{A} = \mathbf{T} * \hat{\mathbf{x}}^{-1}$ [size check: (6x6)x(6x6) = (6x6)]

0.145583796	0.056494192	0.118028215	0.082458173	0.29206849	0.037502758
0.017041581	0.108698522	0.060613517	0.090388933	0.010529423	0.00411795
0.075421171	0.067911827	0.125246063	0.092996306	0.143475694	0.167512317
0.046547862	0.261087645	0.207677165	0.127507786	0.130387719	0.045665122
0.040023371	0.023891235	0.064017388	0.019410444	0.041822476	0.057283624
0.101129613	0.047254488	0.023293963	0.186716883	0.220035426	0.026031326

Step 5 Calculate $\mathbf{I}$

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

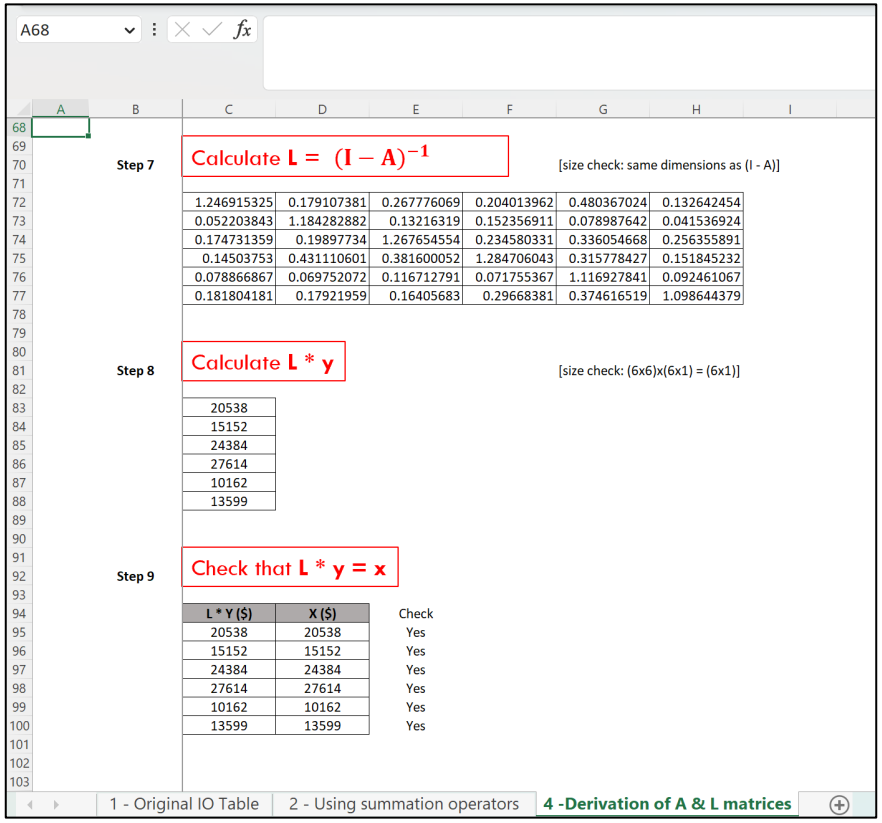
Step 6 Calculate $(\mathbf{I} - \mathbf{A})$ [size check: (6x6) - (6x6) = (6x6)]

0.854416204	-0.056494192	-0.118028215	-0.082458173	-0.29206849	-0.037502758
-0.017041581	0.891301478	-0.060613517	-0.090388933	-0.010529423	-0.00411795
-0.075421171	-0.067911827	0.874753937	-0.092996306	-0.143475694	-0.167512317
-0.046547862	-0.261087645	-0.207677165	0.872492214	-0.130387719	-0.045665122
-0.040023371	-0.023891235	-0.064017388	-0.019410444	0.958177524	-0.057283624
-0.101129613	-0.047254488	-0.023293963	-0.186716883	-0.220035426	0.973968674

Step 7 Calculate $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ [size check: same dimensions as (I - A)]

1 - Original IO Table | 2 - Using summation operators | **4 -Derivation of A & L matrices**

Worked example on Canvas



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Step 7 Calculate $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ [size check: same dimensions as (I - A)]

1.246915325	0.179107381	0.267776069	0.204013962	0.480367024	0.132642454
0.052203843	1.184282882	0.13216319	0.152356911	0.078987642	0.041536924
0.174731359	0.19897734	1.267654554	0.234580331	0.336054668	0.256355891
0.14503753	0.431110601	0.381600052	1.284706043	0.315778427	0.151845232
0.078866867	0.069752072	0.116712791	0.071755367	1.116927841	0.092461067
0.181804181	0.17921959	0.16405683	0.29668381	0.374616519	1.098644379

Step 8 Calculate **L * y** [size check: (6x6)x(6x1) = (6x1)]

20538
15152
24384
27614
10162
13599

Step 9 Check that **L * y = x**

L * Y ($)	X ($)	Check
20538	20538	Yes
15152	15152	Yes
24384	24384	Yes
27614	27614	Yes
10162	10162	Yes
13599	13599	Yes

1 - Original IO Table | 2 - Using summation operators | **4 -Derivation of A & L matrices**

Tutorial

Download the Excel practice activity from Canvas and work through the following questions:

a) Diagonalise the total output (**x**) vector.

b) Calculate the inverse of the diagonal **x** matrix.

c) Using the inverse of diagonal **x** matrix, calculate the **A** matrix. **This is your A matrix.**

d) Construct an identity matrix which is the same size as the **A** matrix.

e) Now calculate **I-A** (**Identity** matrix minus the **A** matrix).

f) Calculate the inverse of the **I-A** matrix. **This is your Leontief inverse L.**

Now examine the fundamental input-output formula $\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y}$ by undertaking the following calculations:

g) Calculate the output required to satisfy the first column of final demand.

h) Calculate the output required to satisfy the second column of final demand.

i) Add your answers for steps g) and h). What do you get? Is your result same as the total output?

Homework for Week 4

Homework week 4 (Due: September 9, 2026)

Use your balanced input-output table (14 industry sectors, 3 value-added entities and 3 final demand entities) to complete the following:

1. **Construct the A-Matrix for your Input-Output System using the matrix techniques discussed in the lecture.**
2. **Construct the L-Matrix for your Input-Output System using the matrix techniques discussed in the lecture.**
3. **Use the fundamental Input-output equation to calculate the output required to satisfy each of the different final demand vectors in your input-output table.**

For example, for the final demand vector of household consumption (let's call it $\mathbf{y}_{HH}$) calculate $\mathbf{x}_{HH}$ using the fundamental input-output formula that was presented in the lecture. Carry out this step for all 3 final demand vectors.

4. **Now that you have the output vectors required to satisfy the demand for each of the final demand vectors, add all these output vectors. Does the result of this addition equal the total output x of your IO system?**

EXPLANATION: Explain the concept of the direct requirements matrix and total requirements matrix in simple words (100 words). Why is your answer in Q4 equal to the total output (100 words)?

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- Matrix diagonalisation
- Matrix inversion
- Direct requirements matrix – direct inputs required to produce $1 of total output *
- Total requirements matrix – total inputs required to produce $1 of total output *
- Fundamental input-output equation

More to come:

- Connecting input-output tables to satellite accounts
- Calculating footprints

* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.